TUMOR LOCALIZATION · CHIP-DESIGN STUDY

Frequency Reduction

How narrow can the measurement band go before localization degrades?
CNN classification · raw S-parameters · LOSO · measured A3 phantom data

1–5 GHz

matches full-band accuracy

1/2

the bandwidth of the full sweep

100%

F5 accuracy retained (full array)

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MOTIVATION

Smaller bandwidth, simpler chip

An integrated version of this system lives or dies on front-end complexity: a narrower sweep means simpler synthesizers, antennas and calibration. The question is what accuracy that costs, and where the band should sit.

1	**WHERE does the information live?**	slide a 2 GHz window across 0.1–8 GHz; full 4-antenna array
2	**HOW WIDE must the band be?**	expand 2 → 3 → 4 → 5 GHz around the winning region
3	**Does it combine with reduced hardware?**	winning band × fewer antennas / reflection-only
4	**Is the band choice robust?**	repeat the whole sweep with a single antenna (S_{11} only)

Every cell: CNN classifier, raw mag+phase input, 100-epoch leave-one-session-out, per-position vote.

STEP 1 · BAND PLACEMENT

A 2 GHz window slid across the sweep

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
0.1 – 2	2 GHz	99.4	96.2	90.9
2 – 4	2 GHz	**99.4**	**97.4**	**92.9**
4 – 6	2 GHz	99.4	89.1	71.7
6 – 8	2 GHz	99.4	97.4	69.7
0.1 – 8 (full)	7.9 GHz	99.3	97.4	100.0

All 16 S-parameters. LOSO per-position vote accuracy (%). Highlighted row = best 2 GHz window (mean across phantoms).

2-4 GHz is the highest-scoring 2 GHz window. Above 4 GHz, F5 drops to ~70%. The empty phantom scores 99.4 in every window.

STEP 2 · BANDWIDTH

Expanding the window around 2-4 GHz

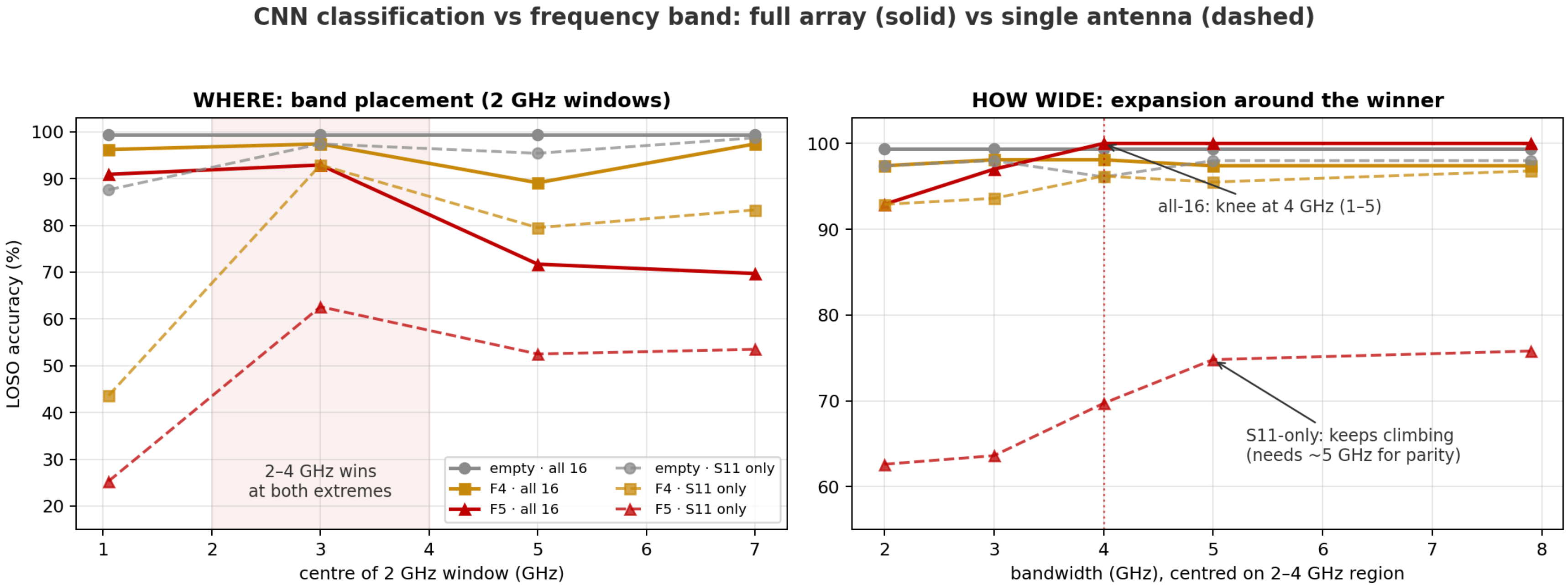
Band (GHz)	Width	Empty	A3 + F4	A3 + F5
2 – 4	2 GHz	99.4	97.4	92.9
1.5 – 4.5	3 GHz	99.4	98.1	97.0
1 – 5	4 GHz	**99.4**	**98.1**	**100.0**
0.5 – 5.5	5 GHz	99.4	97.4	100.0
0.1 – 8 (full)	7.9 GHz	99.3	97.4	100.0

All 16 S-parameters. Windows centred on the winning 2-4 GHz region.

1-5 GHz (4 GHz bandwidth, 3 GHz centre) matches full-band accuracy on all three phantoms, including 100% on F5. 0.5-5.5 GHz scores the same; 1.5-4.5 and 2-4 GHz score 3-7 points lower on F5.

THE FULL PICTURE

Placement and width, at both hardware extremes



Solid = all 16 S-parameters · dashed = single antenna (S11 only). Left: where the 2 GHz window sits. Right: accuracy vs bandwidth around 2-4 GHz.

STEP 3 · COMBINED REDUCTIONS

The 1-5 GHz band with reduced hardware

Hardware @ 1-5 GHz	Empty	A3 + F4	A3 + F5	F5 @ full band
4 antennas, reflection only	**100.0**	**97.4**	**94.9**	*100.0*
2 antennas (1 & 3), full S	99.3	92.9	92.9	*96.0*
2 antennas, reflection only	99.3	92.9	84.8	*90.9*
1 antenna (S11)	96.1	96.2	69.7	*75.8*

With the full array, restricting to 1-5 GHz left accuracy unchanged; with reduced antennas the same restriction scores 3-6 points lower on F5 than full band. 4 antennas reflection-only at 1-5 GHz: 94.9 on F5.

STEP 4 · ROBUSTNESS CHECK

The same window sweep with one antenna

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
0.1 – 2	2 GHz	87.6	43.6	25.3
2 – 4	2 GHz	**97.4**	**92.9**	**62.6**
4 – 6	2 GHz	95.4	79.5	52.5
6 – 8	2 GHz	98.7	83.3	53.5
0.5 – 5.5	5 GHz	98.0	95.5	74.8
0.1 – 8 (full)	7.9 GHz	98.0	96.8	75.8

S11 only. Note 0.1-2 GHz: fine with the full array (90.9 on F5), catastrophic alone (25.3).

With one antenna, 2-4 GHz is again the highest 2 GHz window. F5 accuracy rises with width through ~5 GHz (62.6 at 2 GHz -> 74.8 at 5 GHz). The 0.1-2 GHz window scores 25.3 on F5 with one antenna vs 90.9 with the full array.

RECOMMENDATION

Chip band specification

Spec: 1-5 GHz

4 GHz bandwidth, 3 GHz centre. Matched full-band accuracy on all three phantoms with the full array.

Fallback: 1.5-4.5 GHz

Scored 1-3 points below full band on F5; 2-4 GHz scored 7 below.

Low-scoring regions

F5: 71.7 / 69.7 at 4-6 / 6-8 GHz (full array); 25.3 at 0.1-2 GHz with one antenna.

Combined reductions at 1-5 GHz

F5: 94.9 (4-ant refl) / 92.9 (2-ant) / 84.8 (2-ant refl) / 69.7 (1 antenna).

All results: CNN classification, raw S-parameters, leave-one-session-out, measured A3 phantom data. Details in results/ and docs/CHANGELOG.md.

New constraint, new question: where does it fail?

Follow-up task: stay entirely below 4 GHz (easier chip) and push the bandwidth down until the model breaks. That needs a definition of 'broken':

DEGRADED	mean below 90% of that phantom's full-band accuracy
UNSTABLE	fold-to-fold sigma of 10 points or more: works one session, fails the next
BROKEN	mean below 50%: wrong more often than right (the headline break)
DEEP	mean below 25%: approaching uselessness (still ~10x chance)

Method: descend bandwidth 3 -> 2 -> 1 -> 0.5 -> 0.25 -> 0.1 -> 0.05 GHz with the best placement at every width, plus a nine-window placement scan at 0.25 GHz. All below 4 GHz.

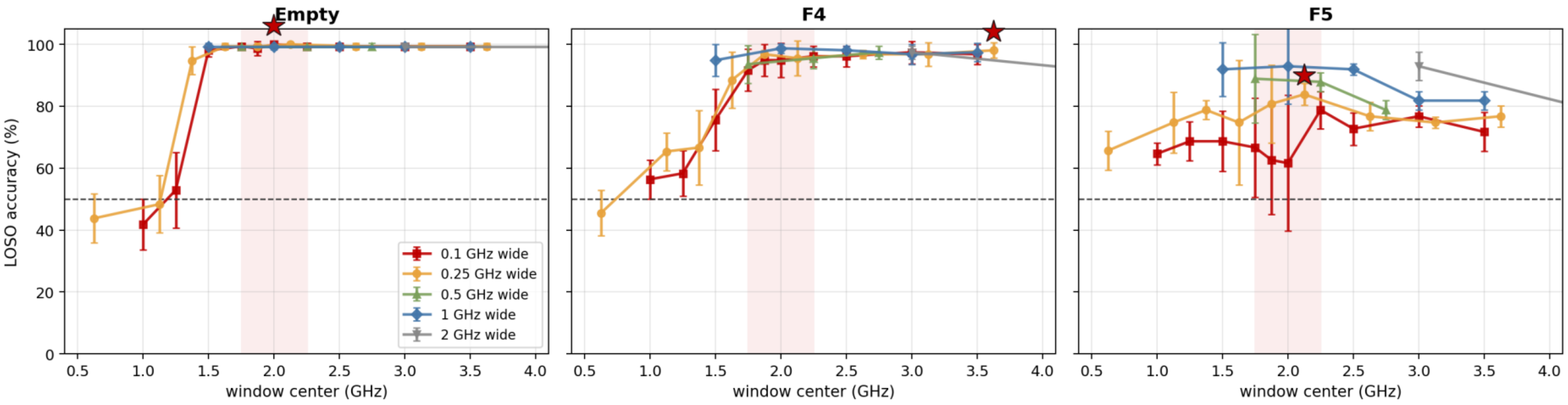
Nine 0.25 GHz windows across the span

Window (GHz)	Empty	A3 + F4	A3 + F5
0.5 - 0.75	43.8 ± 7.9	45.5 ± 7.4	65.7 ± 6.3
1 - 1.25	48.4 ± 9.3	65.4 ± 6.1	74.7 ± 9.7
1.25 - 1.5	94.8 ± 4.5	66.7 ± 12.0	78.8 ± 3.0
1.5 - 1.75	99.3 ± 1.1	88.5 ± 9.0	74.7 ± 20.2
1.75 - 2	99.3 ± 1.1	96.8 ± 1.3	80.8 ± 12.6
2 - 2.25	**100.0 ± 0.0**	**95.5 ± 5.7**	**83.8 ± 3.5**
2.5 - 2.75	99.3 ± 1.1	96.8 ± 1.3	76.8 ± 4.6
3 - 3.25	99.3 ± 1.1	96.8 ± 3.8	74.7 ± 1.7
3.5 - 3.75	99.3 ± 1.1	98.1 ± 2.5	76.8 ± 3.5

Highest window: 2-2.25 GHz (highlighted), with lower fold sigma than 1.75-2. In the red rows (below ~1.25 GHz) empty and F4 score below 50%.

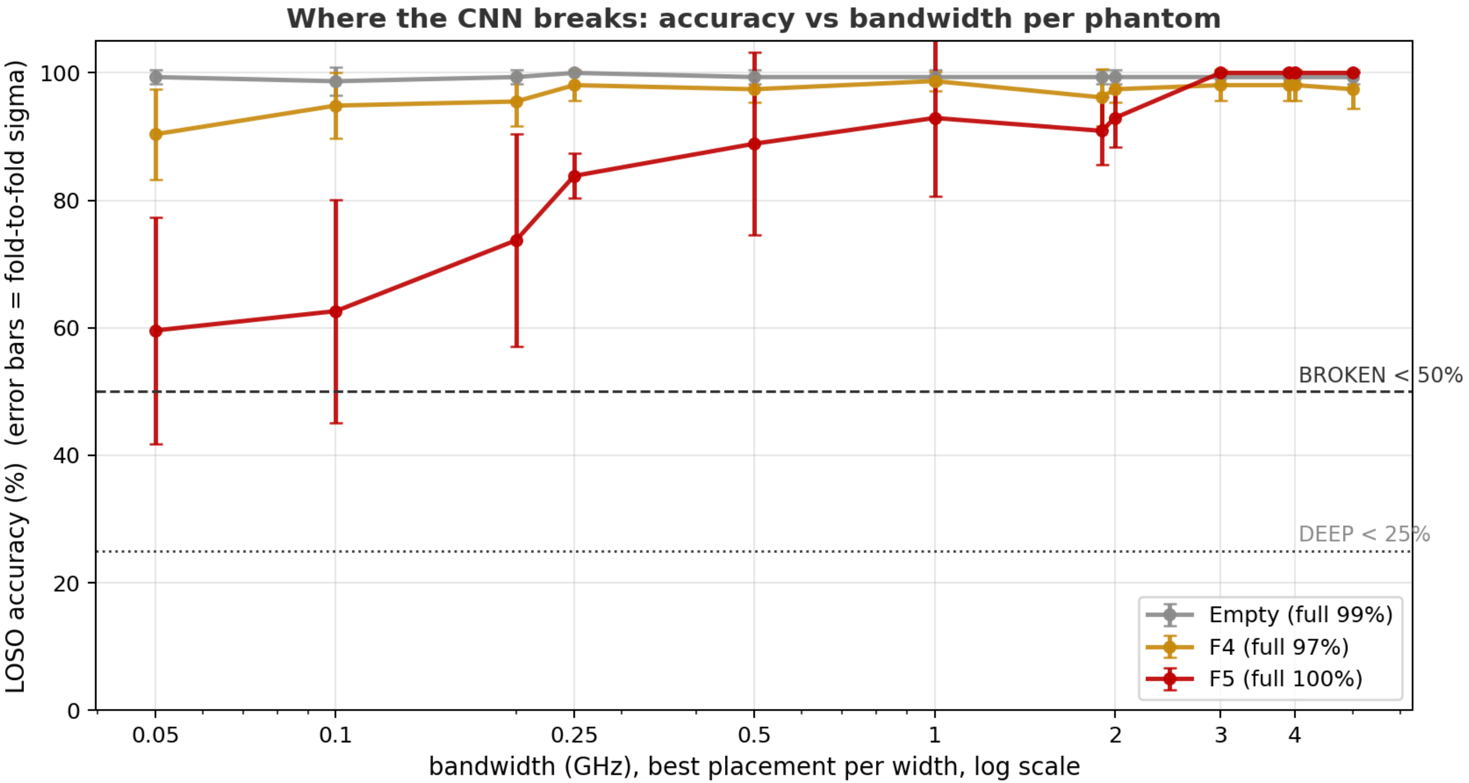
Each phantom has a different best frequency

Where the information lives: train on ONLY one window, slide it across the spectrum. Star = best narrow slot per phantom (each is different)



The model is trained on only one window at a time, slid across the spectrum. Best narrow slot per phantom: empty ~2.0 GHz (99+ everywhere above 1.45), F4 ~3.0 GHz, F5 ~2.2-2.3 GHz. At 100 MHz width F5 scores 78.8 at 2.2-2.3 vs 62.6 at 1.825-1.925, the slot used in the Part 2 descent.

Best placement at every width, down to 0.05 GHz

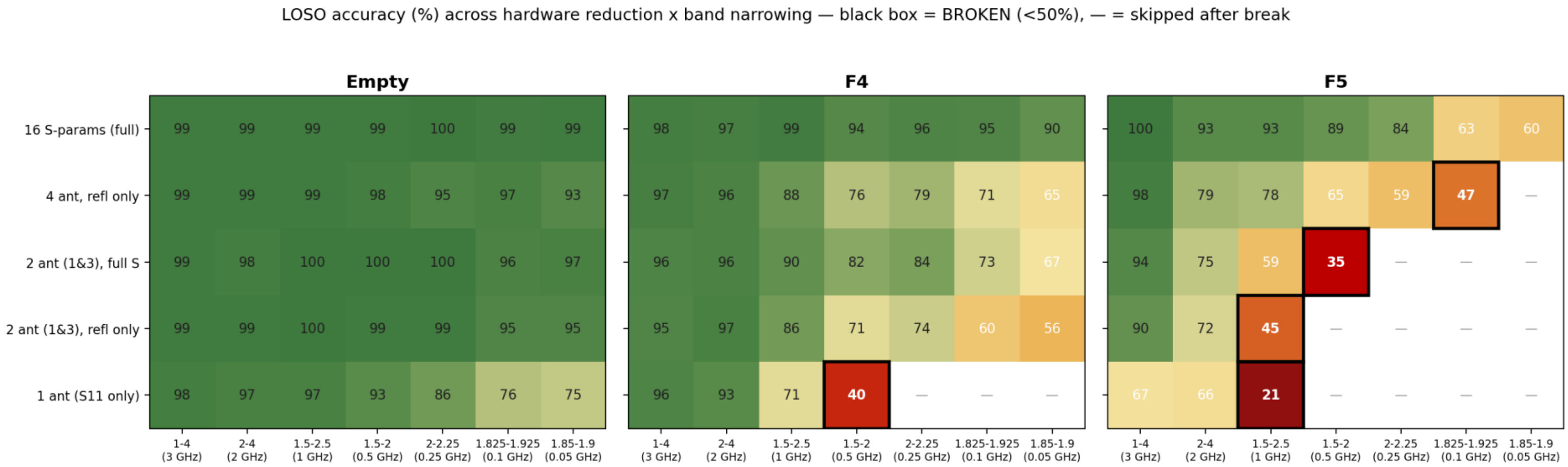


Where each phantom breaks

Failure tier	Empty	A3 + F4	A3 + F5
Degraded (<90% of full band)	never	never	**0.5 GHz**
Unstable (fold sigma >= 10)	never	never	**1 GHz**
Broken (<50%)	never	never	**never**
Floor at 0.05 GHz (50 MHz)	99.3 ± 1.1	90.4 ± 7.1	**59.6 ± 17.8**

With all 16 S-parameters, no phantom scores below 50% at any width: at 50 MHz the floors are 99.3 / 90.4 / 59.6. F5 crosses the degraded tier at 0.5 GHz and the unstable tier at 1 GHz. Part 3 repeats the descent with reduced hardware.

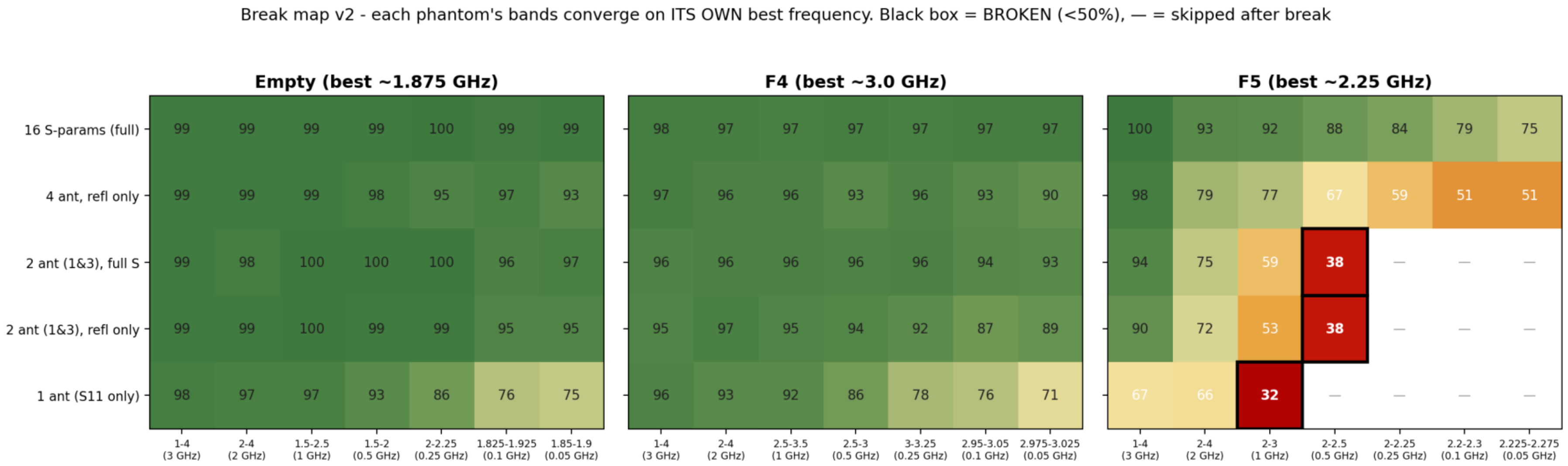
The full break map: 5 hardware levels x 7 bands



Bands narrow left to right at the best placement per width; once a phantom scores below 50% (black box) it stops descending at that hardware level. Blank cells were skipped after the break. Empty does not cross 50% in any cell.

PART 3 · BREAK MAP V2

Same descent with per-phantom band centers



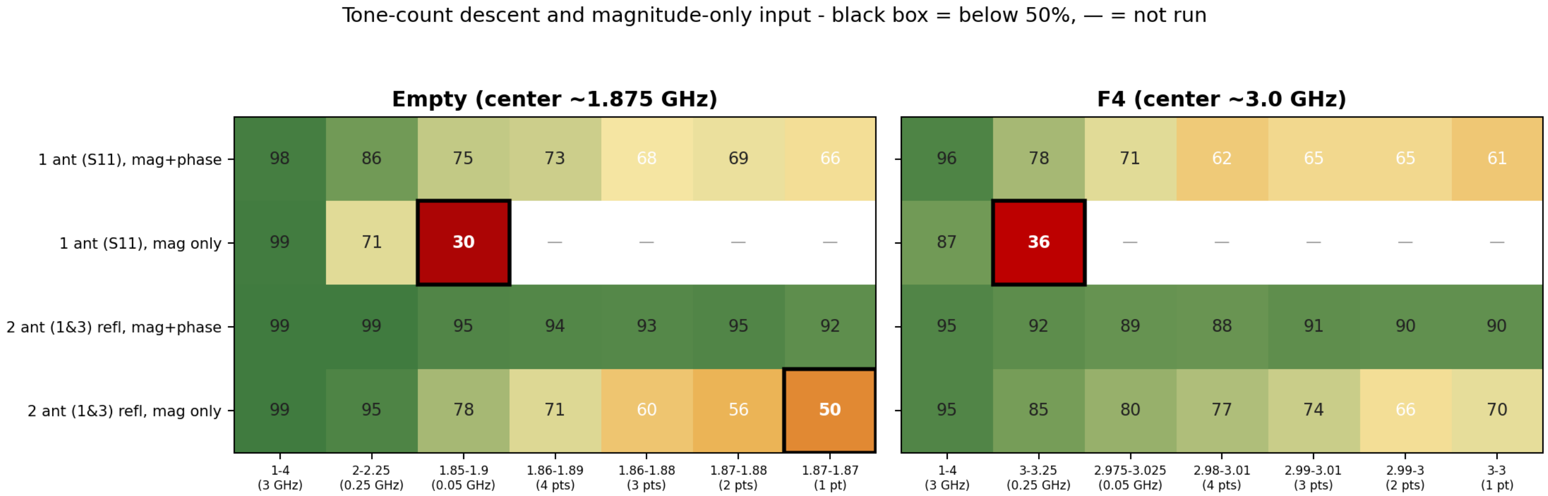
Bands re-centered per phantom: F4 -> 3.0 GHz, F5 -> 2.25 GHz. Empty keeps its original ~1.875 GHz bands: its full-array scan scored 99-100 in every window, and at single antenna the original slot scored higher (76/75 vs 71/59 at 0.1/0.05 GHz). With re-centered bands, F4's single-antenna row no longer crosses 50% (70.5 at 50 MHz vs 39.7 at 0.5 GHz before) and F5's sub-50% cells shift one column narrower.

First band below 50%, original descent

Hardware	Empty	A3 + F4	A3 + F5
16 S-params (full array)	never	never	never (floor 60)
4 antennas, reflection only	never	never	**0.1 GHz (47%)**
2 antennas (1 & 3), full S	never	never	**0.5 GHz (35%)**
2 antennas (1 & 3), reflection only	never	never	**1 GHz (45%)**
1 antenna (S11 only)	never (floor 75)	**0.5 GHz (40%)**	**1 GHz (21%)**

- Empty never scores below 50% in any cell. F4 crosses 50% only at 1 antenna + 0.5 GHz (and not at all with re-centered bands). F5 crosses 50% at every reduced-hardware level.
- F5's first sub-50% band widens as hardware is reduced: 0.1 GHz (4 ant refl) -> 0.5 GHz (2 ant) -> 1 GHz (2 ant refl, 1 ant).

Tone-count descent and magnitude-only input



Configurations that had not crossed 50%, reduced further. Tone columns: 4 -> 1 frequency points on the native 10 MHz grid (1 pt = a single CW tone). Mag-only rows use |S| only (no phase). Single-tone, mag+phase: empty 66.0 (1 ant) / 91.5 (2 ant refl); F4 60.9 / 90.4. Mag-only with one antenna: 87-99 at 1-4 GHz but below 50% by 0.05-0.25 GHz (30.1 / 35.9); with two antennas: 49.7 (empty) and 69.9 (F4) at a single tone.